MEDICAL INSURANCE EXPENDITURE ANALYSIS: AN EMPIRICAL DIAGNOSTIC & MODELING STUDY

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ABSTRACT

In this project, I set out to model individual medical insurance charges (n = 1,338) and investigate how well standard linear regression captures complex, real-world health data. While a baseline Multiple Linear Regression model provided a reasonable starting point (R^2 = 0.742), diagnostic checks revealed two major issues: residual non-normality with heavy tails and heteroscedasticity caused by unmodeled interaction effects—specifically between smoking status, BMI, and age. To address these structural gaps, I engineered interaction features (BMI × Smoker and Age × Smoker) and trained a Random Forest Regressor. This diagnostic-driven approach improved the out-of-sample R^2 from 0.7838 to 0.8632 on the held-out test set, lowering the Test RMSE to $4,608.29. The project highlights how residual diagnostics can guide feature engineering and lead to substantially better predictive performance.

1. PROJECT MOTIVATION & INTRODUCTION

Medical insurance pricing relies heavily on understanding how personal attributes like age, physical health, and lifestyle influence overall medical costs. In this project, my goal was to analyze a dataset of n = 1,338 individuals to see how effectively standard linear regression can predict medical charges, test whether classical regression assumptions actually hold on real-world health data, and evaluate how much model performance improves when introducing non-linear feature interactions and ensemble learning methods.

2. DATASET OVERVIEW & KEY VARIABLES

The dataset contains 1,338 observations across 6 potential covariates and a continuous target variable representing annual medical charges (Y).

Summary Statistics of Numerical Covariates:

- Age (X1): Ranges from 18 to 64 years (Mean = 39.21, Std = 14.05).
- Body Mass Index (X2): Ranges from 15.96 to 53.13 (Mean = 30.66, Std = 6.10), following an approximately symmetric distribution centered near 30.
- Number of Children (X3): Discrete counts from 0 to 5 (Mean = 1.09).
- Smoking Status: Categorical binary indicator (1,064 non-smokers, 274 smokers).
- Sex & Region: Demographic categorical variables (676 male, 662 female; 4 geographic regions).
- Medical Charges (Y): Target variable showing significant right skewness (Mean = $13,270.42, Median = $9,382.03, Max = $63,770.43).

Categorical predictors were dummy-encoded with standard reference baselines.

3. BASELINE LINEAR REGRESSION & RESIDUAL DIAGNOSTICS

I initially split the data into an 80% training set (n = 1,070) and a 20% test set (n = 268). To establish a baseline, I fitted a standard Multiple Linear Regression model using statsmodels.

Baseline Model Estimation:

Y_hat = -11,730 + 257.10*(Age) + 335.63*(BMI) + 427.59*(Children) + 23,660*(Smoker_Yes) - 17.26*(Sex_Male) + Regional Terms

Key Observations from Baseline Fit:

1. Primary Drivers: Smoking status had the single strongest impact (beta = 23,660, $p < 0.001$), followed by Age ($p < 0.001$) and BMI ($p < 0.001$). Sex and Region variables were statistically insignificant ($p > 0.10$).
2. Multicollinearity Check: Computing Variance Inflation Factors (VIF) confirmed no severe collinearity among features, as all values were safely below 1.70 (e.g., VIF_age = 1.02, VIF_bmi = 1.11, VIF_smoker = 1.01).

Baseline OLS Fit Metrics (Training Set):

- R-squared: 0.7420 | Adjusted R-squared: 0.7400
- F-statistic: 380.7 (p-value < 0.001)

Diagnostic Violations Identified:

When I ran residual diagnostics on the fitted linear model, two major assumption violations became apparent:

- Residual Non-Normality: Both the Kernel Density Estimation (KDE) plot and the Normal Q-Q plot exhibited distinct heavy-tail behavior at both extreme quantiles (Jarque-Bera statistic = 612.87, $p < 0.001$). This confirmed that the residuals violate the classical Gaussian noise assumption.
- Heteroscedasticity & Structural Patterns: Plotting fitted values against residuals revealed distinct non-random clustering rather than uniform scatter. This indicated that a simple additive linear model systematically fails to capture the true joint risk profiles in high-cost categories.

4. FEATURE ENGINEERING & MODEL COMPARISONS

To address the non-linear patterns observed during diagnostics, I hypothesized that certain variables act interactively rather than purely additively. Specifically, exploratory scatterplots suggested that being a smoker amplifies the impact of both BMI and Age on medical costs far more than being a non-smoker.

Features Created:

Age^2, BMI^2, Age × Smoker, BMI × Smoker

Cross-Validation Strategy:

I evaluated both an expanded Linear Regression model and a Random Forest Regressor using 5-Fold Cross-Validation on the training data (p = 12):

Model Architecture	Feature Space	5-Fold Mean CV R^2
Linear Regression	Baseline + Interaction Terms	0.8362
Random Forest Regressor	Ensembled Trees (max_depth=5)	0.8569

5. FINAL EVALUATION & KEY LEARNINGS

Evaluating the final Random Forest model on the completely unseen test set (n = 268) yielded strong out-of-sample performance:

- Baseline OLS Test R^2: 0.7838
- **Final Ensembled Test R^2: 0.8632**
- **Final Test RMSE: $4,608.29**

Key Reflections:

1. Diagnostics reveal hidden structure: Running baseline residual checks (Q-Q plots, residual vs. fitted plots) was critical. It clearly showed where the linear model was failing, which directly guided my feature engineering strategy.
2. Interaction terms matter in health data: Linear models assume each variable acts independently, but in health insurance, smoking status multiplies the risk associated with BMI and Age. Adding explicit interaction terms allowed linear regression to improve from 0.742 to 0.836 in R^2.
3. Tree models capture non-linear thresholds naturally: Random Forest performed best because tree-based splits naturally handle threshold boundaries (like high BMI combined with smoking) without imposing rigid linearity constraints.